

---

**Question #1 of 120**

Question ID: 1212005

Questions 1 through 18 relate to Ethical and Professional Standards. (27 minutes)

Which of the following statements made in a marketing brochure is a violation of the Standards?

- A) "Roger Langley, Chartered Financial Analyst, has been a portfolio manager for ten years and passed all three levels of the CFA examinations on his first attempts."
- B) "Jennifer York has passed the Level II exam and will earn the right to use the CFA designation after completing the Level III exam this June."
- C) "Paul Yeng, CFA, has retired from the firm after 25 years of service. Much of the firm's past success can be attributed to Yeng's efforts as an analyst and portfolio manager."

Explanation

According to Standard VII(B) Reference to the CFA Institute, the CFA Designation, and CFA Program, citing an expected date for completing a level of the CFA Program is a misuse of the CFA designation. (Module 3.9)

**For Further Reference:**

(Study Session 1, Module 3.9, LOS, 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

---

**Question #2 of 120**

Question ID: 1212006

Hedge Funds Unlimited, a global hedge fund, has publicly acknowledged in writing that it has adopted the CFA Institute Code and Standards as its policies. Which of the following is *least likely* a violation of the firm's policies?

- A) An analyst at the firm working overseas uses material nonpublic information as allowed by local law to make investment decisions for discretionary client accounts.
- B) A junior analyst at the firm uses a subscription to his local newspaper and the opinions of his friends and colleagues to make investment recommendations for discretionary client accounts.
- C) A CFA candidate at the firm, who is registered for the Level III exam, includes reference to participation in the CFA program and her status as a Level III candidate in her biographical background.

Explanation

All investment personnel in this example are subject to the CFA Institute Code and Standards as part of the firm's established policies. The candidate's reference to her Level III status and the inclusion of such information in her biographical information is not in violation of the CFA Institute Code and Standards. Candidates may clearly reference their participation in the CFA program, provided such reference does not imply the achievement of any type of partial designation. The analyst is considered a candidate since she is registered to take the next scheduled examination. The Code and Standards prohibit using material nonpublic information. Since the Code and Standards are *stricter* than the local law, they must be followed by the analyst. The junior analyst failed to exercise diligence and thoroughness in making investment recommendations and failed to have a reasonable and adequate basis for such recommendations. (Modules 3.1, 3.3, 3.7, and 3.9)

**For Further Reference:**

(Study Session 1, Modules 3.1, 3.3, 3.7, and 3.9, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

---

**Question #3 of 120**

Question ID: 1212001

According to the Code and Standards, members and candidates who are involved in distributing an initial public offering (IPO) of equity shares and wish to participate in the IPO:

- A) may participate unless the IPO is oversubscribed.**
- B) may not participate because this creates a conflict of interest.**
- C) must obtain pre-clearance from a supervisor before participating.**

Explanation

Standard VI(B) Priority of Transactions recommends, but does not require, that a member or candidate obtain pre-clearance from his or her supervisor before participating in an equity IPO. Guidance for Standard III(B) Fair Dealing states that members and candidates distributing IPO shares must distribute shares in an oversubscribed IPO to clients and may not withhold shares for themselves. (Modules 3.4, 3.8)

**For Further Reference:**

(Study Session 1, Modules 3.4,3.8, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

---

**Question #4 of 120**

Question ID: 1211997

Linda Bryant, CFA, is an employee of Roomkin Investment House, which underwrites equity and debt offerings. She has been approached by SimthCo to consult on a private debt placement. According to CFA Institute Standards of Professional Conduct, before Bryant agrees to accept this job, she is required to:

- A) obtain written consent from Roomkin after submitting details of the arrangement.**
- B) talk to her immediate supervisor and get her approval to take this consulting job.**

- C)** inform SimthCo in writing that she will accept the job and provide details of the arrangement to Roomkin in writing.

Explanation

To comply with Standard IV(B) Additional Compensation Arrangements, Bryant must obtain written consent from her employer before undertaking the independent consulting project. Bryant must also provide a description of the types of services being provided, the length of time the arrangement will last, and the compensation she expects to receive for her services. (Module 3.6)

**For Further Reference:**

(Study Session 1, Module 3.6, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

---

**Question #5 of 120**

Question ID: 1211999

To comply with the Code and Standards, analysts who send research recommendations to clients must:

- A) keep records of all the data and analysis that went into creating the report.**
- B)** send recommendations only to those clients for whom the investments are suitable.
- C)** not send recommendations without including the underlying analysis and basic investment characteristics.

Explanation

Standard V(C) Record Retention requires members to maintain records of the data and analysis they use to develop their research recommendations. Recommendations may be brief, in capsule form, or simply a list of buy/sell recommendations. A list of recommendations may be sent without regard to suitability, including both safe income stocks and aggressive growth stocks, for example. (Module 3.7)

**For Further Reference:**

(Study Session 1, Module 3.7, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

---

**Question #6 of 120**

Question ID: 1211998

Amy Brooks, a Level III CFA candidate, has been given supervisory responsibilities. In carrying out her responsibilities, Brooks has discovered that the firm's compliance system is inadequate. She informed her supervisor, who is not supportive of Brooks's efforts to correct the situation. According to CFA Institute Standards of Professional Conduct, Brooks:

- A)** has satisfied her obligation under the Code and Standards by informing her manager of the situation.

- B)** must dissociate herself from the firm if the firm is not in compliance with the CFA Institute Standards.
- C) should decline in writing to accept supervisory responsibilities until an adequate compliance system is adopted.**

Explanation

Standard IV(C) Responsibilities of Supervisors indicates that a member should decline supervisory responsibility in writing until the firm adopts reasonable compliance procedures. Otherwise, Brooks cannot adequately exercise her responsibility. (Module 3.6)

**For Further Reference:**

(Study Session 1, Module 3.6, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

---

**Question #7 of 120**

Question ID: 1211996

Not including the results of terminated accounts when calculating historical performance is recommended by:

- A)** both GIPS and the Standard concerning performance presentation.
- B)** GIPS, but not by the Standard concerning performance presentation.
- C) neither GIPS nor the Standard concerning performance presentation.**

Explanation

Standard III(D) Performance Presentation recommends that terminated accounts be included in historical performance calculations. (Module 3.5)

**For Further Reference:**

(Study Session 1, Module 3.5, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

---

**Question #8 of 120**

Question ID: 1212000

Ken Toma, CFA, has just completed an extensive analysis and concluded that the demand for vacation rentals in Hawaii will far exceed the supply for the foreseeable future. Toma writes a research report stating, "Based on the fact that the demand for Hawaiian beach vacations will exceed the supply of rooms for the foreseeable future, I recommend the purchase of shares of The Hawaiian REIT, a diversified portfolio of Hawaiian beachfront resorts." If Toma presents this report to his clients, he will *most likely* violate the CFA Institute Standards by:

- A) not distinguishing between fact and opinion.**
- B)** not considering the suitability of the investment for his clients.

**C)** failing to have a reasonable and adequate basis for his recommendation.

#### Explanation

Standard V(B) Communication with Clients and Prospective Clients requires that Toma separate opinion from fact. Toma's statement that excess demand will persist into the foreseeable future is an opinion, not a fact. Toma has established a reasonable basis for his recommendation through his analysis. Suitability does not become an issue until a client chooses to act on Toma's recommendation. (Module 3.7)

#### **For Further Reference:**

(Study Session 1, Module 3.7, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

---

### **Question #9 of 120**

Question ID: 1211993

Derek Stevens, CFA, manages the pension plan assets of Colors, Inc. When voting proxies for plan equities, Stevens owes a fiduciary duty to:

- A)** the plan trustees who hired him.
- B) the plan participants and beneficiaries.**
- C)** the managers, stockholders, and bondholders of Colors, Inc., equally.

#### Explanation

Under Standard III(A) Loyalty, Prudence, and Care, the fiduciary duty in this case is to plan participants and beneficiaries, not shareholders or plan trustees. (Module 3.4)

#### **For Further Reference:**

(Study Session 1, Module 3.4, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

---

### **Question #10 of 120**

Question ID: 1211994

An analyst at Romer changes her rating on TelSky from "buy" to "hold" and sends an email explaining the change to all clients and firm brokers. Subsequently, Paul Stevens, CFA, a broker at Romer, receives a call from a client who wants to buy 15,000 shares of TelSky. Stevens must:

- A) advise his client of the change in recommendation before accepting the order.**
- B)** not accept the order until the customer has had time to receive and read the new report.
- C)** accept the order without mentioning the ratings change because the order is unsolicited.

### Explanation

According to Standard III(B) Fair Dealing, if a client places an order that goes against the firm's recommendation for that security, members and candidates should inform the client of the discrepancy between the order and the firm's recommendation before accepting the order. (Module 3.4)

### **For Further Reference:**

(Study Session 1, Module 3.4, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

---

## **Question #11 of 120**

Question ID: 1207399

Which of the following is one of the nine major sections of the GIPS standards?

- A) Verification.
- B) Private equity.
- C) Sub-advisers.

### Explanation

Private equity is one of the nine major sections of the GIPS standards; the others are not.

### **For Further Reference:**

(Study Session 1, Module 5.1, LOS 5.d)

CFA® Program Curriculum, Volume 1, page 239

---

## **Question #12 of 120**

Question ID: 1211995

Edie Pschorr, CFA, notices that a bond is priced at 98.0 in one market and 98.4 in another market. Pschorr places an order to buy a large number of these bonds in the first market and simultaneously places an order to sell the same number of bonds in the second market. The bond's price increases to 98.2 in the first market and decreases to 98.2 in the second market. Are Pschorr's trades a violation of the Code and Standards?

- A) No.
- B) Yes, because they violate the Standard concerning fair dealing.
- C) Yes, because they violate the Standard concerning market manipulation.

### Explanation

The trader has carried out an arbitrage transaction. Because she did not exhibit any intent to distort prices or trading volume, the member did not violate Standard II(B) Market Manipulation. Standard III(B) Fair Dealing is concerned with fair treatment of clients and is not relevant to this transaction. (Modules 3.3, 3.4)

### **For Further Reference:**

Study Session 1, Modules 3.3, 3.4, LOS 3.a, 3.b, 3.c

### Question #13 of 120

Question ID: 1212002

Greg Hoffman, CFA, has been offered a cash payment by Hill Manufacturing, Inc. to write a research report on their company. According to the Code and Standards, Hoffman:

- A) must disclose the nature of the compensation from Hill in his research report.**
- B) may not accept compensation from Hill to produce research on the company.**
- C) may produce the research report but may not make a recommendation on Hill's securities.**

#### Explanation

Standard I(B) Independence and Objectivity requires disclosure of the nature of any compensation from the subject company. Standard VI(A) Disclosure of Conflicts more generally requires disclosure of any potential conflict of interest in research reports and investment recommendations. (Modules 3.1, 3.8)

#### **For Further Reference:**

Study Session 1, Modules 3.1, 3.8, LOS 3.a, 3.b, 3.c

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

---

### Question #14 of 120

Question ID: 1212003

Rhonda Morrow, CFA, is an analyst for Waller & Madison, a brokerage and investment banking firm. Waller & Madison is a market maker for CorpEast, and Tim Waller, a principal in Morrow's firm, sits on CorpEast's board. Morrow has been asked to write a research report on CorpEast. According to the Standard regarding disclosure of conflicts, Morrow:

- A) must not write the report.**
- B) must disclose that Waller & Madison is a market maker in CorpEast shares but not that Waller is a board member.**
- C) may write the report if she discloses both that Waller & Madison is a market maker in CorpEast shares and that Waller sits on the CorpEast board.**

#### Explanation

To comply with Standard VI(A) Disclosure of Conflicts, both the market-making activities by the firm and the directorship held by a principal in the firm must be disclosed. (Module 3.8)

#### **For Further Reference:**

Study Session 1, Module 3.8, LOS 3.a, 3.b, 3.c

CFA® Program Curriculum, Volume 1, page 46

### Question #15 of 120

Question ID: 1211991

John Farr, CFA, has accumulated several pieces of nonmaterial nonpublic information about CattleCorp from his contacts with the company. From analysis based on this information, together with public information, Farr concludes that CattleCorp will have unexpectedly low earnings this year. Farr has contacted the company, but they will not confirm his conclusion. According to CFA Institute Standards of Professional Conduct, Farr:

- A) may trade or make recommendations based on his analysis.**
- B) may not trade or make recommendations based on his analysis.
- C) may trade or make recommendations based on his analysis only if his company's compliance officer determines that the nonpublic information he used was not material.

#### Explanation

According to Standard II(A) Material Nonpublic Information, Farr is free to act under the mosaic theory because nonmaterial nonpublic information does not fall within the prohibition on trading based on material nonpublic information. He should keep detailed documentation of his analysis to document that he did not advise or act based on material nonpublic information. (Module 3.3)

#### **For Further Reference:**

Study Session 1, Module 3.3, LOS 3.a, 3.b, 3.c

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

---

### Question #16 of 120

Question ID: 1207400

According to the GIPS standards, which of the following statements is *most accurate*?

- A) Firms are required to obtain independent third-party verification for a claim of GIPS compliance.
- B) GIPS compliant firms are required to maintain written documentation of policies and procedures used to establish and maintain compliance with GIPS.**
- C) To initially claim compliance with GIPS, a company must present a minimum of ten years (or since the firm's inception if less than ten years) of GIPS-compliant performance data.

#### Explanation

GIPS require firms that claim compliance to maintain written documentation of their policies and procedures for complying with GIPS. Verification of GIPS compliance is optional. In order to initially claim compliance with GIPS, a firm must have a minimum of five years (or since firm inception) of GIPS-compliant data. After the first compliant presentation, another year of compliant performance must be added each year until the compliant performance history reaches at least ten years.

#### **For Further Reference:**

## Question #17 of 120

Question ID: 1212004

Shan Ang, CFA, is a portfolio manager at Huang Investments. Lian Jan, an old friend of Ang's, is an executive recruiter in the same city. Jan proposes that she will refer any high-level executives that she places locally to Ang, in exchange for one round of golf at Ang's country club for each new client. According to the Standard concerning referral fees, Ang would be required to disclose this referral arrangement:

- A) only to all prospective clients referred by Jan.
- B) to his employer and all prospective clients referred by Jan.**
- C) to all prospective clients, current clients, and his employer.

### Explanation

Standard VI(C) Referral Fees states that members and candidates must disclose to employers and to affected prospects and clients, before entering into any formal agreement for services, any benefits received for the recommendation of services provided by the member. (Module 3.8)

### **For Further Reference:**

Study Session 1, Module 3.8, LOS 3.a, 3.b, 3.c

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

---

## Question #18 of 120

Question ID: 1211992

Yvette Michaels, CFA, an analyst for Torborg Investments, inadvertently overhears a conversation between two executives of Collective Healthcare in which they mention an upcoming tender offer for Network, a stock she covers. Michaels has followed both companies extensively and feels their consolidation would be very beneficial for both companies. She tells her supervisor, a senior analyst, about the proposed tender offer. Michaels' actions are:

- A) in violation of the Standards.**
- B) not in violation of the Standards because she told only her supervisor.
- C) not in violation of the Standards because she has not traded shares of Network or changed her report on the company.

### Explanation

Michaels has violated Standard II(A) Material Nonpublic Information. Members who possess material nonpublic information are prohibited from acting or causing others to act on that information. She may not share the information with anyone except designated supervisory or compliance employees within her firm. Disclosing to her supervisor, who is not identified as a designated supervisor of compliance issues, is not permitted. (Module 3.3)

**For Further Reference:**

Study Session 1, Module 3.3, LOS 3.a, 3.b, 3.c

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 46

CFA® Program Curriculum, Volume 1, page 55

---

**Question #19 of 120**

Question ID: 1207402

**Questions 19 through 30 relate to Quantitative Methods. (18 minutes)**

The odds for an event occurring are calculated by dividing:

- A) one by the probability that the event occurs.
- B) the probability that the event does not occur by the probability that an event occurs.
- C) the probability that the event occurs by the probability that the event does not occur.

Explanation

If  $p$  is the probability that an event occurs, then the odds for the event occurring are expressed as  $p / (1 - p)$ , or the probability that the event occurs divided by the probability that the event does not occur. The odds against the event are expressed as the reciprocal of the odds for the event.

**For Further Reference:**

(Study Session 2, Module 8.1, LOS 8.c)

CFA® Program Curriculum, Volume 1, page 460

---

**Question #20 of 120**

Question ID: 1207401

Which of the following statements about return distributions is *most accurate*?

- A) With positive skewness, the median is greater than the mean.
- B) If skewness is positive, the average magnitude of positive deviations from the mean is smaller than the average magnitude of negative deviations from the mean.
- C) If a return distribution has positive excess kurtosis and the analyst uses statistical models that do not account for the fatter tails, the analyst will underestimate the likelihood of extreme outcomes.

Explanation

If a return distribution has positive excess kurtosis, statistical models that do not account for the fatter tails will underestimate the likelihood of very bad or very good outcomes. A distribution with positive skewness will have a mean greater than the median and larger average positive deviations than average negative deviations.

**For Further Reference:**

(Study Session 2, Module 7.3, LOS 7.j, 7.k, 7.l)

CFA® Program Curriculum, Volume 1, page 424

**Question #21 of 120**

Question ID: 1207411

An analyst plans to use the following test statistic:

$$t_{n-1} = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

This test statistic is appropriate for a hypothesis about:

- A) the mean difference of two normal populations.
- B) the population mean of a normal distribution with unknown variance.**
- C) the equality of two population means of two normally distributed populations based on independent samples.

Explanation

When testing hypotheses about the population mean, the sample standard deviation must be used in the denominator of the test statistic when the population standard deviation is unknown, the population is normal, and/or the sample is large. The statistic is a *t*-stat with  $n - 1$  degrees of freedom. The numerator is the sampling error for the population mean if the true mean is  $\mu_0$  and the denominator is the standard error of the sample mean around the true mean.

**For Further Reference:**

(Study Session 3, Module 11.2, LOS 11.g)

CFA® Program Curriculum, Volume 1, page 620

---

**Question #22 of 120**

Question ID: 1207403

Which of the following statements about covariance and the correlation coefficient is *least accurate*?

- A) Covariance is a measure of how the returns of two assets tend to move together over time.
- B) The correlation coefficient is computed by dividing the covariance of returns on two assets by the individual variances of returns for the two assets.**
- C) The covariance of returns between two assets is equal to the correlation between the returns of the two assets, multiplied by the product of their standard deviations of returns.

Explanation

Dividing the covariance between returns of two assets by the individual *standard deviations* of returns of the two assets yields the correlation coefficient.

**For Further Reference:**

(Study Session 2, Module 8.2, LOS 8.k)

CFA® Program Curriculum, Volume 1, page 480

---

## Question #23 of 120

Question ID: 1207408

Greg Goldman, research analyst in the fixed-income area of an investment bank, needs to determine the average duration of a sample of twenty 15-year fixed-coupon investment grade bonds. Goldman first categorizes the bonds by risk class and then randomly selects bonds from each class. After combining the bonds selected (bond ratings and other information taken as of March 31 of the current year), he calculates a sample mean duration of 10.5 years. Assuming that the actual population mean duration is 9.7 years, which of the following statements about Goldman's sampling process and sample is *least accurate*?

- A) Goldman is using time-series data.
- B) The sample mean is a random variable.
- C) The sampling error is 0.8 years.

### Explanation

The data are cross-sectional, which means that it is a sample of observations taken at a single point in time. Time-series data are observations taken at specific and equally spaced points in time (for example, the monthly returns on a specific stock for the period January 1 through December 31 of a given year). The sampling error is the difference between a sample statistic (here, the mean) and the corresponding population parameter, or  $10.5 - 9.7 = 0.8$ . The sample statistic (here, the mean) is itself a random variable and has its own probability distribution.

### **For Further Reference:**

(Study Session 3, Module 10.1, LOS 10.d)

CFA® Program Curriculum, Volume 1, page 573

---

## Question #24 of 120

Question ID: 1212007

An investment with an initial cost of \$30,000 is sold for \$60,000 after two years. The annual return on a continuously compounded basis is *closest to*:

- A) 35%.
- B) 38%.
- C) 40%.

### Explanation

The two-year holding period return is +100%. The effective annual rate is  $(1 + 1.00)^{1/2} - 1 = 41.42\%$ . The equivalent annual rate of interest on a continuously compound basis is  $\ln(1.4142) = 34.66\%$ . Alternatively, we can calculate the two-year continuously compounded rate as  $\ln(2) = 69.31\%$  and divide by two to get 34.66%.

### **For Further Reference:**

(Study Session 3, Module 9.3, LOS 9.o)

CFA® Program Curriculum, Volume 6, page 342

CFA® Program Curriculum, Volume 6, page 388

---

## Question #25 of 120

Question ID: 1207404

If Stock X has a standard deviation of returns of 18.9% and Stock Y has a standard deviation of returns equal to 14.73% and returns on the stocks are perfectly positively correlated, the standard deviation of an equally weighted portfolio of the two is:

- A) 10.25%.
- B) 14.67%.
- C) **16.82%.**

Explanation

The standard deviation of two stocks that are perfectly positively correlated is the weighted average of the standard deviations:  $0.5(18.9) + 0.5(14.73) = 16.82\%$ . This relationship is true only when the correlation is one. Otherwise, you must use the formula:

$$\sigma_p = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2 \rho_{1,2}}$$

**For Further Reference:**

(Study Session 2, Module 8.3, LOS 8.I)

CFA® Program Curriculum, Volume 1, page 480

---

**Question #26 of 120**

Question ID: 1207405

An investment manager wants to select three analysts from a group of six analysts to receive first-, second-, and third-place awards for outstanding performance. In how many ways can the investment manager make the three awards?

- A) 20 ways.
- B) 54 ways.
- C) **120 ways.**

Explanation

When order of selection matters, as it does here, use the permutation formula:

$${}_n P_r = \frac{n!}{(n-r)!}$$

$${}_6 P_3 = 6! / (6-3)! = (6 \times 5 \times 4 \times 3 \times 2 \times 1) / (3 \times 2 \times 1) = (6 \times 5 \times 4) = 120$$

**For Further Reference:**

(Study Session 2, Module 8.3, LOS 8.o)

CFA® Program Curriculum, Volume 1, page 495

---

**Question #27 of 120**

Question ID: 1207409

Which of the following statements about the central limit theorem is *least accurate*?

- A) The standard deviation of the sample mean is called the standard error of the sample mean.
- B) **The standard error of the sample mean can be estimated by dividing the population standard deviation by  $\sqrt{n-1}$ .**

- C)** The sample means for large sample sizes will have an approximately normal distribution regardless of the distribution of the underlying population.

Explanation

The standard error of the sample mean can be estimated by dividing the population standard deviation by  $\sqrt{n}$ .

**For Further Reference:**

(Study Session 3, Module 10.1, LOS 10.e, 10.f)

CFA® Program Curriculum, Volume 1, page 576

CFA® Program Curriculum, Volume 1, page 576

---

**Question #28 of 120**

Question ID: 1207410

An investment analyst takes a random sample of 100 aggressive equity funds and calculates the average beta as 1.7. The sample betas have a standard deviation of 0.4. Using a 95% confidence interval and a z-statistic, which of the following statements about the confidence interval and its interpretation is *most accurate*? The analyst can be confident at the 95% level that the interval:

- A)** 1.580 to 1.820 includes the mean of the population beta.
- B) 1.622 to 1.778 includes the mean of the population beta.**
- C)** 1.634 to 1.766 includes the mean of the population beta.

Explanation

Given that the population variance is unknown and the sample size is large, the 95% confidence interval for the population mean is:

$$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$$

The confidence interval is:

$$1.7 \pm 1.96 \left( \frac{0.4}{\sqrt{100}} \right) = 1.7 \pm 1.96 (0.04) = 1.7 \pm 0.0784 = 1.622 \text{ to } 1.778$$

**For Further Reference:**

(Study Session 3, Module 10.2, LOS 10.j)

CFA® Program Curriculum, Volume 1, page 581

---

**Question #29 of 120**

Question ID: 1207412

Which is the correct test statistic for a test of the null hypothesis that a population variance is equal to a chosen value?

- A)** F-statistic.
- B)** t-statistic.
- C) Chi-square statistic.**

Explanation

This is a test of the value of a single variance and is based on a test statistic with a performed via the chi-square distribution.

**For Further Reference:**

(Study Session 3, Module 11.3, LOS 11.j)

CFA® Program Curriculum, Volume 1, page 636

---

**Question #30 of 120**

Question ID: 1207406

A random variable follows a continuous uniform distribution over the range 2 to 12. The probability of an outcome greater than 5 but less than or equal to 8 is *closest to*:

- A) 27%.
- B) 30%.
- C) 36%.

Explanation

The probability is  $(8 - 5) / (12 - 2) = 30\%$ . For a continuous distribution there is no difference between "less than" and "less than or equal to" because the probability of any specific outcome is effectively zero.

**For Further Reference:**

(Study Session 3, Module 9.1, LOS 9.h)

CFA® Program Curriculum, Volume 6, page 355

CFA® Program Curriculum, Volume 6, page 366

---

**Question #31 of 120**

Question ID: 1212008

**Questions 31 through 42 relate to Economics. (18 minutes)**

To benefit from price discrimination, a monopolist *least likely* needs to have:

- A) a higher-quality product at a premium price and a lower-quality alternative.
- B) a way to prevent reselling between types of consumers.
- C) two identifiable groups of consumers with different price elasticities of demand for the product.

Explanation

Price discrimination involves a single product, not two alternatives. As long as the company faces a downward-sloping demand curve, can identify at least two groups of customers with different price elasticities of demand, and can prevent reselling between groups, the company can profit from price discrimination.

**For Further Reference:**

(Study Session 4, Module 13.4, LOS 13.b, 13.d)

CFA® Program Curriculum, Volume 2, page 64

CFA® Program Curriculum, Volume 2, page 64

---

### Question #32 of 120

Question ID: 1207421

Automatic stabilizers are government programs that require no legislation and tend to:

- A) automatically increase spending at the same growth rate as real GDP.
- B) reduce interest rates, thus stimulating aggregate demand.
- C) **change the government budget deficit in an opposite direction to economic growth.**

#### Explanation

Automatic stabilizers are built-in features that tend to automatically promote a budget deficit during a recession and a budget surplus during an inflationary boom, without a change in policy.

#### **For Further Reference:**

(Study Session 5, Module 16.3, LOS 16.o)

CFA® Program Curriculum, Volume 2, page 300

---

### Question #33 of 120

Question ID: 1212009

An economy is in full-employment equilibrium. If the government unexpectedly decreases the tax rate, the economy is *most likely* to experience:

- A) **an increase in employment in the short run.**
- B) a decrease in the price level in the short run.
- C) no change in employment in the short run.

#### Explanation

Short-run equilibrium may occur above full employment, for example as a result of an increase in aggregate demand caused by a decrease in taxes. Both employment and the price level increase in the short run. Above-full employment causes upward pressure on wages that will reduce short-run aggregate supply until, in the long run, output returns to its full-employment level with a still-higher equilibrium price level.

#### **For Further Reference:**

(Study Session 4, Module 14.3, LOS 14.k)

CFA® Program Curriculum, Volume 2, page 161

---

### Question #34 of 120

Question ID: 1207418

Pat Bannerman is analyzing economic indicators to form an opinion on whether an economic contraction has ended. Which of the following turning points should Bannerman *most appropriately* interpret as a coincident indicator suggesting economic growth is entering the early stage of a new expansion?

- A) **Real personal income has begun increasing.**
- B) The unemployment rate has begun decreasing.
- C) Building permits for new houses have begun increasing.

### Explanation

Real personal income is a coincident indicator with turning points that tend to coincide with business cycle turning points. The unemployment rate is a lagging indicator, here suggesting an expansion has been underway for some time. Building permits are a leading indicator because builders may seek permits in anticipation of an economic expansion that has not begun yet.

### **For Further Reference:**

(Study Session 4, Modules 15.1, 15.2, LOS 15.a, 15.d, 15.i)

CFA® Program Curriculum, Volume 2, page 198

CFA® Program Curriculum, Volume 2, page 219

CFA® Program Curriculum, Volume 2, page 237

---

## **Question #35 of 120**

Question ID: 1207424

Three years ago, the U.S. dollar/euro exchange rate was 1.32 USD/ EUR. Over the last three years, the price level in the United States has increased by 18%, and the price level in the eurozone has increased by 12%. If the current exchange rate is 1.40 USD/EUR, the real exchange rate over the period has:

- A) increased, and eurozone goods are now more expensive to U.S. consumers.**
- B) decreased, and eurozone goods are now more expensive to U.S. consumers.**
- C) increased, and U.S. goods are now more expensive to eurozone consumers.**

### Explanation

For the base period, three years ago, the real exchange rate is the same as the nominal exchange rate, 1.32 USD/EUR. The real exchange rate over the period has changed from 1.32 to  $1.40 \times (112 / 118) = 1.3288$ . This increase in the real USD/EUR exchange rate indicates that the base currency (EUR) has appreciated in real terms, so that eurozone goods are now more expensive in real terms to U.S. consumers.

### **For Further Reference:**

(Study Session 5, Module 18.1, LOS 18.a)

CFA® Program Curriculum, Volume 2, page 397

---

## **Question #36 of 120**

Question ID: 1207422

Which of the following conditions is *most likely* to result in expansionary effects from fiscal policy being felt when an economic expansion is already underway?

- A) Slow economic growth is being caused by supply shortages rather than low aggregate demand.**
- B) Expansionary fiscal policy requires policymakers to recognize an economic contraction and enact legislation.**
- C) Government borrowing to finance expansionary spending is increasing interest rates faced by private sector borrowers.**

### Explanation

Fiscal policy takes effect with a lag due to the time it takes for policymakers to recognize the business cycle stage (recognition lag) and enact legislation to change fiscal policy (action lag), and the time it takes for individuals and businesses to react to the policy changes (impact lag). As a result of these lags, fiscal policy changes may result in pro-cyclical rather than countercyclical effects. Supply shortages and higher interest rates as a result of government borrowing (the crowding-out effect) both act to reduce the expansionary effect of expansionary fiscal policy but are not related directly to its time lags.

**For Further Reference:**

(Study Session 5, Module 16.3, LOS 16.r)

CFA® Program Curriculum, Volume 2, page 316

---

**Question #37 of 120**

Question ID: 1207413

Which of the following is *most likely* a characteristic of monopolistic competition?

- A) Producer decisions are interdependent.
- B) Each producer offers a differentiated product.**
- C) Producers face horizontal demand curves.

Explanation

Differentiated products are a feature of monopolistic competition markets. Interdependence is a characteristic of oligopoly markets. Horizontal demand curves facing producers are a feature of perfect competition.

**For Further Reference:**

(Study Session 4, Module 13.1, LOS 13.a)

CFA® Program Curriculum, Volume 2, page 64

---

**Question #38 of 120**

Question ID: 1207419

The Fisher effect describes the relationship among:

- A) savings, investment, the fiscal balance, and the trade balance.
- B) credit expansion, investor expectations, and the business cycle.
- C) nominal interest rates, real interest rates, and expected inflation.**

Explanation

The Fisher effect states that a nominal interest rate is the sum of a real interest rate and the expected inflation rate.

**For Further Reference:**

(Study Session 5, Module 16.1, LOS 16.e)

CFA® Program Curriculum, Volume 2, page 272

---

**Question #39 of 120**

Question ID: 1207415

Which of the following events is *most likely* to increase short-run aggregate supply (shift the curve to the right)?

- A) Inflation that results in an increase in goods prices.
- B) High unemployment puts downward pressure on money wages.**
- C) An increase in government spending intended to increase real output.

Explanation

Falling money wages would cause businesses to increase (profit-maximizing) output levels at each price level for final goods and services. Changes in the price level of goods and services are represented by a movement along a short-run aggregate supply curve, not a shift in the curve. A rise in resource prices will decrease aggregate supply. An increase in government spending will shift the aggregate demand curve but not the aggregate supply curve.

**For Further Reference:**

(Study Session 4, Module 14.2, LOS 14.h)

CFA® Program Curriculum, Volume 2, page 148

---

**Question #40 of 120**

Question ID: 1207417

An unexpected increase in businesses' inventory-to-sales ratios is *most likely* to occur as an economy:

- A) reaches a trough.
- B) enters a contraction phase.
- C) approaches the peak of an expansion.**

Explanation

At the end or peak of an expansion, economic activity begins to slow, sales are less than planned, and excess inventory accumulates, increasing inventory-to-sales ratios. To reduce inventory-to-sales ratios to desired levels, firms decrease production, which is one of the causes of a contraction.

**For Further Reference:**

(Study Session 4, Module 15.1, LOS 15.b)

CFA® Program Curriculum, Volume 2, page 200

---

**Question #41 of 120**

Question ID: 1207423

Costs of reducing restrictions on international trade are *most likely* to be borne by domestic producers:

- A) as a whole.
- B) that lack an absolute advantage.
- C) that lack a comparative advantage.**

Explanation

While increased international trade has benefits that outweigh its costs, those costs are borne by domestic producers in industries that compete with imports and lack a comparative advantage. Domestic producers that have a comparative advantage relative to foreign producers are likely to gain from increased trade.

**For Further Reference:**

(Study Session 5, Module 17.1, LOS 17.b, 17.c)

CFA® Program Curriculum, Volume 2, page 341

CFA® Program Curriculum, Volume 2, page 343

---

**Question #42 of 120**

Question ID: 1207420

A change from a neutral monetary policy to a contractionary monetary policy is *most likely* to be reflected in the economy by:

- A) decreases in price for financial assets.**
- B) decreased lending rates in the banking system.
- C) depreciation of the domestic currency in the foreign exchange market.

Explanation

Contractionary monetary policy typically results in increased interest rates, which decrease the prices of financial assets by decreasing the present value of future cash flows. Increasing interest rates typically cause the domestic currency to appreciate.

**For Further Reference:**

(Study Session 5, Module 16.2, LOS 16.i)

CFA® Program Curriculum, Volume 2, page 283

---

**Question #43 of 120**

Question ID: 1212010

**Questions 43 through 60 relate to Financial Reporting and Analysis. (27 minutes)**

Peri, Inc. has entered into a long-term contract to build a warehouse that includes a bonus payment of 10% of the contract value, to be paid if the building is completed by a specific date. In determining the revenue to recognize, the firm should calculate the proportion of the performance obligations fulfilled to date, and multiply that by the contract value:

- A) not including the bonus.**
- B) including the bonus if it is probable they will receive it.
- C) including the bonus only if they are certain they will receive it.**

Explanation

Revenue recognition standards require that firm conclude that they will not have to reverse any cumulative revenue recognized, as they would have to if the bonus is not received. If nonreversal is only probable, they should not include the bonus in calculating revenue.

**For Further Reference:**

(Study Session 7, Module 21.2, LOS 21.c)

CFA® Program Curriculum, Volume 3, page 79

---

**Question #44 of 120**

Question ID: 1207431

An analyst gathered the following data about a company:

- Collections from customers are \$5,000.
- Depreciation is \$800.
- Cash expenses (including taxes) are \$2,000.
- Tax rate = 30%.
- Net cash increased by \$1,000.

If inventory increases over the period by \$800, cash flow from operations equals:

- A) \$1,600.**
- B) \$2,400.**
- C) \$3,000.**

#### Explanation

Use the direct method.

|                            |                 |
|----------------------------|-----------------|
| Collections from customers | \$5,000         |
| Cash expenses              | <u>-\$2,000</u> |
| Cash flow from operations  | \$3,000         |

Cash expenses are given. If you had been given COGS, you would need to adjust that for inventory changes to get cash expenses for inputs. Depreciation is a non-cash change.

Changes in depreciation are used with the indirect method. Net change in cash will reflect CFI and CFF, not just CFO.

#### **For Further Reference:**

(Study Session 7, Module 23.2, LOS 23.f)

CFA® Program Curriculum, Volume 3, page 199

---

## **Question #45 of 120**

Question ID: 1207434

Greene Company discloses that its net income for the most recent period was reduced by a writedown of inventory to net realizable value. What effect is the inventory writedown *most likely* to have on Greene's net income in future periods?

- A) Increase.**
- B) Decrease.**
- C) No effect.**

#### Explanation

In future periods, lower-valued inventory will result in lower cost of sales and higher net income.

#### **For Further Reference:**

(Study Session 8, Module 25.4, LOS 25.h)

CFA® Program Curriculum, Volume 3, page 334

---

## Question #46 of 120

Question ID: 1207438

Compared to reporting in a country where life insurance payments on key employees are deductible for tax, a company that makes such payments and reports in a country in which they are not tax deductible would report:

- A) a lower statutory tax rate.
- B) a higher effective tax rate.**
- C) a greater deferred tax asset.

### Explanation

In the case where insurance premiums paid are not tax deductible, taxes would be higher so the effective tax rate (tax expense / pretax income) would be higher. Expenses on the income statement that are not deductible for tax and will not reverse are permanent differences and will not affect deferred tax items.

### **For Further Reference:**

(Study Session 8, Module 27.5, LOS 27.i)

CFA® Program Curriculum, Volume 3, page 471

---

## Question #47 of 120

Question ID: 1207432

Which of the following is *most likely* presented on a common-size balance sheet or common-size income statement?

- A) Total asset turnover.
- B) Operating profit margin.**
- C) Return on common equity.

### Explanation

Operating profit margin can be read directly from a common-size income statement. Asset turnover and return on equity mix balance sheet and income statement items.

### **For Further Reference:**

(Study Session 7, Modules 21.5, 24.2, 24.3, LOS 21.i, 21.j, 24.b)

CFA® Program Curriculum, Volume 3, page 109

CFA® Program Curriculum, Volume 3, page 109

CFA® Program Curriculum, Volume 3, page 256

---

## Question #48 of 120

Question ID: 1207429

An analyst gathers the following data about a company:

- The company had 1 million shares of common stock outstanding for the entire year.
- The company's beginning stock price was \$50, its ending price was \$70, and its average price was \$60.
- The company had 100,000 warrants outstanding for the entire year. Each warrant allows the holder to buy one share of common stock at \$50 per share.

How many shares of common stock should the company use in computing its diluted earnings per share?

- A) 1,100,000.
- B) 1,083,333.
- C) 1,016,667.

Explanation

Use the Treasury stock method:

- Step 1:** Determine the number of common shares created if the warrants are exercised = 100,000.
- Step 2:** Calculate the cash inflow if the warrants are exercised:  $(100,000)(\$50 \text{ per share}) = \$5,000,000$ .
- Step 3:** Calculate the number of shares that can be purchased with these funds using the average market price (\$60 per share):  $5,000,000 / 60 = 83,333$  shares.
- Step 4:** Calculate the net increase in common shares outstanding from the exercise of the warrants:  
 $100,000 - 83,333 = 16,667$ .
- Step 5:** Add the net increase in common shares from the exercise of the warrants to the number of common shares outstanding for the entire year:  $1,000,000 + 16,667 = 1,016,667$ .

**For Further Reference:**

(Study Session 7, Module 21.4, LOS 21.g)

CFA® Program Curriculum, Volume 3, page 99

---

**Question #49 of 120**

Question ID: 1207430

Under U.S. GAAP, land owned by the firm is *most likely* to be reported on the balance sheet at:

- A) historical cost.
- B) fair market value minus selling costs.
- C) historical cost less accumulated depreciation.

Explanation

Unless impairment has been recognized, land is reported at historical cost and is not subject to depreciation. Increases in value are not reflected in balance sheet values under U.S. GAAP.

**For Further Reference:**

(Study Session 7, Module 22.4, LOS 22.e)

CFA® Program Curriculum, Volume 3, page 133

---

**Question #50 of 120**

Question ID: 1207425

Which of the following items is *least likely* to contain details about various accruals, adjustments, balances, and management assumptions?

- A) Income statement.
- B) Supplementary schedules.

**C)** Discussion and analysis by management.

Explanation

The income statement reports the amounts for each of the major line items within the general categories of revenues and expenses. The various accruals, adjustments, and management assumptions are implicit in the reported amounts but are not specifically explained in the income statement. Much of the detail contained in various accruals, adjustments, and management assumptions that go into the financial statements can be found in the footnotes to the statements and Management's Discussion and Analysis. Supplementary schedules contain additional information, including a more detailed breakdown of certain large account balances.

**For Further Reference:**

(Study Session 6, Module 19.2, LOS 19.c)

CFA® Program Curriculum, Volume 3, page 24

---

**Question #51 of 120**

Question ID: 1207439

A permanent difference between pretax and taxable income is *least likely* to arise when a firm:

- A)** receives tax-exempt interest.
- B)** uses the installment sales method for financial reporting.
- C)** pays premiums on life insurance of key employees.

Explanation

The installment sales method of revenue recognition does not result in permanent differences between pretax and taxable income. Premium payments on life insurance of key employees is an expense on the financial statements, but is not deducted on tax returns. Tax exempt interest is recognized as revenue on the financial statements. These items result in permanent differences between pretax income and taxable income.

**For Further Reference:**

(Study Session 8, Module 27.5, LOS 27.f)

CFA® Program Curriculum, Volume 3, page 462

---

**Question #52 of 120**

Question ID: 1207426

According to the IASB Conceptual Framework for Financial Reporting, what are the two fundamental qualitative characteristics of financial statements?

- A)** Timeliness and comparability.
- B)** Verifiability and understandability.
- C)** Relevance and faithful representation.

Explanation

The IASB Conceptual Framework for Financial Reporting describes the two fundamental qualitative characteristics of financial statements as relevance and faithful representation. The Conceptual Framework lists timeliness, comparability, verifiability, and understandability as characteristics that enhance relevance and faithful representation.

**For Further Reference:**

(Study Session 6, Module 20.2, LOS 20.c)

CFA® Program Curriculum, Volume 3, page 54

---

**Question #53 of 120**

Question ID: 1207433

A firm uses the first-in first-out (FIFO) cost flow assumption. Compared to gross profit with a periodic inventory system, the firm's gross profit with a perpetual inventory system would be:

- A) lower.
- B) higher.
- C) the same.

Explanation

For a firm using FIFO, gross profit is the same whether the firm uses a periodic or perpetual inventory system. For a firm using LIFO or average cost, gross profit can be different depending on the choice of inventory system.

**For Further Reference:**

(Study Session 8, Module 25.2, LOS 25.c)

CFA® Program Curriculum, Volume 3, page 316

---

**Question #54 of 120**

Question ID: 1207428

Which costs are *least likely* to be reported as an expense in the current accounting period?

- A) Period costs.
- B) Costs of producing inventory.
- C) Loan interest that has not yet been paid.

Explanation

Inventory costs are expensed when items are sold under the matching principle. As an extreme example, if no sales are made, no costs of inventory production are expensed for the period. Period costs are expensed during the period. Under the accrual method, interest accrued during the period is expensed, regardless of whether it has been paid during the period.

**For Further Reference:**

(Study Session 7, Module 21.3, LOS 21.d)

CFA® Program Curriculum, Volume 3, page 84

---

**Question #55 of 120**

Question ID: 1207441

While motivation and opportunity both can lead to low quality of financial reporting, a third important contributing factor is:

- A) poor financial controls.

**B) rationalization of the actions.**

**C) pressure to meet earnings expectations.**

Explanation

A mindset that allows rationalization is the third important condition underlying low-quality financial reporting. Poor financial controls are an example of opportunity and pressure to meet earnings expectations is a possible motivation.

**For Further Reference:**

(Study Session 9, Module 29.1, LOS 29.e)

CFA® Program Curriculum, Volume 3, page 558

---

**Question #56 of 120**

Question ID: 1207442

In a period of rising prices, management of a company that reports under IFRS is *least likely* to attempt to influence analysts' opinions of its financial results by:

**A) liquidating inventory.**

**B) increasing the useful lives of assets.**

**C) emphasizing earnings that exclude nonrecurring costs.**

Explanation

During periods of rising prices, liquidating inventory is a potential way for companies that use the LIFO inventory cost method to report higher current-period earnings, but LIFO is not permitted under IFRS. Managers attempting to influence analysts' opinions may manipulate accounting assumptions or emphasize non-GAAP measures in their financial reports.

**For Further Reference:**

(Study Session 9, Modules 29.1, 29.3, LOS 29.g, 29.i)

CFA® Program Curriculum, Volume 3, page 567

CFA® Program Curriculum, Volume 3, page 591

---

**Question #57 of 120**

Question ID: 1207436

Granite, Inc., owns a machine with a carrying value of \$3.0 million and a salvage value of \$2.0 million. The present value of the machine's future cash flows is \$1.7 million. The asset is permanently impaired. Granite should:

**A) immediately write down the machine to its salvage value.**

**B) immediately write down the machine to its recoverable amount.**

**C) write down the machine to its recoverable amount as soon as it is depreciated down to salvage value.**

Explanation

Under IFRS, when an asset is permanently impaired, it must be written down to its recoverable amount (greater of value in use or fair value less selling costs) in the period in which the impairment is recognized.

**For Further Reference:**

**Question #58 of 120**

Question ID: 1207440

Yamaska Mining issued a 5-year, \$50 million face, 6% semiannual bond when market interest rates were 7%. The market yield of the bonds was 8% at the beginning of the next year. Using the effective interest rate method, what is the initial balance sheet liability, and what is the interest expense that the company should report for the first half of the second year of the bond's life (the third semiannual period)?

|    | <u>Initial liability</u> | <u>Interest expense, first half of year 2</u> |
|----|--------------------------|-----------------------------------------------|
| A) | \$47,920,849             | \$1,689,853                                   |
| B) | \$47,920,849             | \$1,750,000                                   |
| C) | \$50,000,000             | \$1,500,000                                   |

Explanation

This is a discount bond since the market interest rate at issuance exceeds the coupon rate. The initial liability is equal to the proceeds received when the bond was issued. We can find this amount from the following calculation:  $FV = 50,000,000$ ;  $N = 10$ ;  $I = 3.5$ ;  $PMT = 1,500,000$ ;  $CPT \rightarrow PV = \$47,920,848.67$ . Change  $N$  to 8 and calculate  $PV$  to get liability value at the beginning of the second year of the bond's life, 48,281,511. Interest expense for the next semiannual period is  $48,281,511(0.035) = \$1,689,853$ . The subsequent change in the market rate has no effect on the amortization of the discount.

**For Further Reference:**

(Study Session 8, Modules 28.1, 28.2, LOS 28.a, 28.b)

CFA® Program Curriculum, Volume 3, page 490

CFA® Program Curriculum, Volume 3, page 494

**Question #59 of 120**

Question ID: 1207437

Clement Company has revalued an intangible asset with an indefinite life upward by €25 million. In its financial statements, Clement will *most likely*:

- A) disclose how it determined the fair value of the intangible asset.
- B) report lower net income in subsequent periods because of increased amortization expense on the asset.
- C) report higher assets, net income, and shareholders' equity in the most recent period than it would have reported under the cost model.

Explanation

For firms that revalue assets upward, IFRS requires disclosure of the date the asset was revalued, how management determined its fair value, the asset's carrying value using the historical cost model, and (for intangible assets) whether the asset's useful life is finite or indefinite. Although assets and shareholders' equity will increase as a result of the revaluation, net income will not increase. The

increase in the value of the asset is reported as a revaluation surplus in shareholders' equity. Amortization expense will not increase because indefinite-lived intangible assets are not amortized.

**For Further Reference:**

(Study Session 8, Modules 26.2, 26.3, 26.4, LOS 26.g, 26.h, 26.l)

CFA® Program Curriculum, Volume 3, page 407

CFA® Program Curriculum, Volume 3, page 407

CFA® Program Curriculum, Volume 3, page 417

---

**Question #60 of 120**

Question ID: 1207435

As a result of a recent acquisition, Lombard, Inc., has placed the following items on their balance sheet as of the beginning of their fiscal year:

Goodwill     \$30 million

Patent        \$10 million   Expires in 10 years.

Trademark   \$15 million   Expires in 15 years, renewable at minimal cost.

If Lombard amortizes intangible assets using the straight line method, the amortization expense on these assets for the fiscal year will be:

- A) \$1 million.
- B) \$2 million.
- C) \$3 million.

Explanation

Goodwill has an indefinite life and is not amortized. A trademark or other intangible asset that has an expiration date but is renewable at minimal cost is treated as having an indefinite life and is not amortized. The patent has a finite life and its cost will be amortized at the rate of \$1 million each year over ten years under the straight-line method.

**For Further Reference:**

(Study Session 8, Module 26.2, LOS 26.f)

CFA® Program Curriculum, Volume 3, page 406

---

**Question #61 of 120**

Question ID: 1207443

**Questions 61 through 72 relate to Corporate Finance. (18 minutes)**

Responsibilities of a board of directors' nominations committee are *least likely* to include:

- A) recruiting qualified members to the board.
- B) selecting an external auditor for the company.
- C) evaluating the independence of directors.

Explanation

Selecting an external auditor (subject to shareholder approval) is a responsibility of the Board's audit committee.

**For Further Reference:**

(Study Session 10, Module 31.1, LOS 31.f)

CFA® Program Curriculum, Volume 4, page 21

---

**Question #62 of 120**

Question ID: 1207447

The following data applies to LeVeit Company:

- LeVeit has a target debt-to-equity ratio of 0.5.
- LeVeit's bonds are currently yielding 10%.
- LeVeit is a constant growth (5%) firm that just paid a dividend of \$3.00.
- LeVeit's stock sells for \$31.50 per share.
- The company's marginal tax rate is 40%.

The company's weighted after-tax cost of capital is *closest* to:

- A) 10.5%.**
- B) 11.0%.**
- C) 12.0%.**

Explanation

$$k_{ce} = D_1 / P_0 + g = (3)(1.05) / (31.50) + 0.05 = 0.15 \text{ or } 15\%$$

$$k_d(1 - t) = 10\%(1 - 0.4) = 6\%$$

If the company has a debt-to-equity ratio of 0.5, it will have \$0.50 in debt for each \$1.00 in equity.  $V = \text{debt} + \text{equity} = 0.5 + 1 = 1.5$ . Therefore, the weight is 33.3% ( $0.5 / 1.5$ ) for the debt component and 66.7% ( $1.0 / 1.5$ ) for the equity component.

$$\begin{aligned} \text{WACC} &= \frac{D}{V} (k_{\text{debt}}) (1 - t) + \left( \frac{E}{V} \right) k_s \\ &= 0.333 (6.0\%) + 0.677 (15\%) \\ &= 2\% + 10\% = 12\% \end{aligned}$$

**For Further Reference:**

(Study Session 10, Module 33.1, LOS 33.h)

CFA® Program Curriculum, Volume 4, page 88

---

**Question #63 of 120**

Question ID: 1207444

Timely Taxis, Ltd. has signed a long-term lease for 20 underground parking spots at \$150 each per month for its fleet of taxis. The firm currently has 18 taxis in operation and is performing an NPV analysis on the purchase of a 19th taxi. The cost of parking for the 19th taxi is *best* described as:

- A) a sunk cost.**
- B) an opportunity cost.**
- C) an incremental cost.**

### Explanation

The 19th space is neither an incremental cost, nor an opportunity cost or a type of cannibalization. It is a sunk cost since the firm has already committed to parking for 20 taxis. The cost of the 19th parking space is not directly relevant to the capital budgeting decision.

### **For Further Reference:**

(Study Session 10, Module 32.1, LOS 32.b)

CFA® Program Curriculum, Volume 4, page 50

---

## **Question #64 of 120**

Question ID: 1207448

If a firm uses the weighted average cost of capital (WACC) to discount cash flows of higher than average risk projects, which one of the following will *most likely* occur?

- A) Project NPVs will be understated.
- B) The firm will reject profitable projects.
- C) The overall risk of the firm's investments will rise over time.**

### Explanation

Risky projects will seem relatively more attractive than they actually are, causing them to be undertaken a disproportionately high percentage of the times they are considered.

### **For Further Reference:**

(Study Session 10, Module 33.1, LOS 33.a)

CFA® Program Curriculum, Volume 4, page 78

---

## **Question #65 of 120**

Question ID: 1207452

A guarantee stating that a payment will be made upon receipt of goods or services is *best* described as:

- A) commercial paper.
- B) a banker's acceptance.**
- C) a revolving line of credit.

### Explanation

Banker's acceptances are guarantees from a bank on behalf of a firm that has ordered goods, stating that a payment will be made upon receipt of the goods.

### **For Further Reference:**

(Study Session 11, Module 35.1, LOS 35.g)

CFA® Program Curriculum, Volume 4, page 189

---

## **Question #66 of 120**

Question ID: 1207445

Wreathfield, Inc., is choosing between two mutually exclusive projects. The cash flows for the two projects are below. The firm has a cost of capital of 12%, and the risk of the projects is equivalent to the average risk of the firm.

|            | 0       | 1     | 2     | 3     | 4     | 5     | 6     |
|------------|---------|-------|-------|-------|-------|-------|-------|
| Project J: | -12,000 | 4,000 | 5,000 | 6,000 |       |       |       |
| Project K: | -20,000 | 3,000 | 3,000 | 3,000 | 5,000 | 8,000 | 8,000 |

Wreathfield should accept:

- A) Project J.
- B) Project K.
- C) Neither project J nor project K.

#### Explanation

The firm should reject both projects.

|            | 0         | 1       | 2       | 3       | 4       | 5       | 6       |
|------------|-----------|---------|---------|---------|---------|---------|---------|
| Project J: | -\$12,000 | \$4,000 | \$5,000 | \$6,000 |         |         |         |
| Project K: | -\$20,000 | \$3,000 | \$3,000 | \$3,000 | \$5,000 | \$8,000 | \$8,000 |

Project J:  $I/Y = 12$ ;  $CF_0 = (12,000)$ ;  $CF_1 = 4,000$ ;  $CF_2 = 5,000$ ;  $CF_3 = 6,000$ ;  $CPT \rightarrow NPV = -172$ .

Project K: same process.  $NPV = -1,025$ .

#### **For Further Reference:**

(Study Session 10, Module 32.2, LOS 32.d)

CFA® Program Curriculum, Volume 4, page 52

## Question #67 of 120

Question ID: 1207453

If firms Acme and Butler have the same amount of sales and equal quick ratios, but Acme's receivables turnover is higher, it is *most likely* that:

- A) Butler has better liquidity than Acme.
- B) Butler has a lower cash ratio than Acme.
- C) Acme's average days of receivables is higher than Butler's.

#### Explanation

Given that they have the same amount of sales and Acme's receivables turnover (sales/average accounts receivable) is higher, Acme must have lower average accounts receivable than Butler. Given that they have equal quick ratios, subtracting accounts receivable from the numerators of the quick ratios of both firms will produce a cash ratio for Butler that is lower than the cash ratio for Acme.

#### **For Further Reference:**

(Study Session 11, Module 35.1, LOS 35.b)

CFA® Program Curriculum, Volume 4, page 159

## Question #68 of 120

Question ID: 1207451

Fireball Company has fixed operating costs of \$120,000 and fixed financing costs of \$120,000. Fireball's variable costs are \$4 per unit of output. If Fireball sells 200,000 units for \$5 each, it will generate:

- A) a net profit.
- B) an operating loss.
- C) **an operating profit but a net loss.**

### Explanation

Revenue =  $200,000 \times \$5 = \$1,000,000$ . Variable costs =  $200,000 \times \$4 = \$800,000$ . Operating income =  $\$1,000,000 - \$800,000 - \$120,000 = \$80,000$ . Net income =  $\$80,000 - \$120,000 = -\$40,000$ .

Alternatively, operating breakeven quantity =  $\$120,000 / (\$5 - \$4) = 120,000$  units. Breakeven quantity =  $(\$120,000 + \$120,000) / (\$5 - \$4) = 240,000$  units. Unit sales are less than the breakeven quantity but greater than the operating breakeven quantity.

### **For Further Reference:**

(Study Session 11, Module 34.1, LOS 34.d, 34.e)

CFA® Program Curriculum, Volume 4, page 143

CFA® Program Curriculum, Volume 4, page 143

---

## Question #69 of 120

Question ID: 1207449

A firm's optimal capital budget can be found by moving along its investment opportunity schedule until:

- A) it exhausts its capital budget.
- B) average project return is equal to average cost of capital.
- C) **the next project's return is less than the marginal cost of capital.**

### Explanation

The firm would not want to exhaust its capital budget on "bad" projects, (i.e., projects with  $IRR < \text{cost of capital}$  [ $NPV < 0$ ]). They should continue to invest as long as the project's return is greater than the marginal cost of capital of the firm. When the project's  $IRR = \text{cost of capital}$ , the  $NPV = 0$ . Such a project will only make the firm larger; it will add nothing to the stock value. The investment opportunity schedule plots expected project returns from highest to lowest IRR.

### **For Further Reference:**

(Study Session 10, Module 33.1, LOS 33.d)

CFA® Program Curriculum, Volume 4, page 82

---

## Question #70 of 120

Question ID: 1207454

A company's excess cash balances can *most appropriately* be invested in:

- A) common stock.
- B) corporate bonds.

**C) commercial paper.**

Explanation

Excess cash balances should be invested to earn a positive return, but should remain in liquid instruments with relatively stable values. Examples include U.S. Treasury bills, short-term federal agency securities, bank certificates of deposit, banker's acceptances, time deposits, repurchase agreements, commercial paper, money market mutual funds, and adjustable-rate preferred stock.

**For Further Reference:**

(Study Session 11, Module 35.1, LOS 35.d)

CFA® Program Curriculum, Volume 4, page 164

---

**Question #71 of 120**

Question ID: 1207450

A company's schedule of the costs of debt and equity shows that an additional \$3 million of debt can be issued at an after-tax cost of 3%, and additional equity of \$9 million can be issued at a cost of 6%. The company plans to maintain a capital structure of 30% debt and 70% equity. At what level of new capital financing will the marginal cost of capital change with the issuance of new debt?

- A) \$3 million.
- B) \$10 million.
- C) \$13 million.

Explanation

The break point at which new debt will increase the marginal cost of capital is calculated as \$3 million (the level of debt when the cost will change) divided by the percent of debt in the capital structure (30%). Break point = \$3 million / 0.3 = \$10 million in total new capital.

**For Further Reference:**

(Study Session 10, Module 33.2, LOS 33.k)

CFA® Program Curriculum, Volume 4, page 102

---

**Question #72 of 120**

Question ID: 1207446

Compared to the NPV profile of a project with an initial outlay of 1,000 and annual after-tax cash flows of 300 for four years, an alternative project with an initial outlay and annual after-tax cash flows 20% higher would have a Y-intercept (vertical intercept) that is:

- A) unchanged and an X-intercept that is greater.
- B) increased by 20% and an X-intercept that is greater.
- C) increased by 20% and an X-intercept that is unchanged.

Explanation

The vertical intercept is the NPV at a zero interest rate and is 20% greater for a project that is 20% larger. For the \$1,000 project, the vertical intercept is  $(4 \times 300) - 1,000 = 200$ , and for the 20% larger project the vertical intercept is  $1.2 \times [(4 \times 300) - 1,000] = 1.2(200) = 240$ . The horizontal intercept is the project IRR, which is unchanged by equal percentage increases in the initial outlay and after-tax cash flows.

**For Further Reference:**

### Question #73 of 120

Question ID: 1207457

**Questions 73 through 86 relate to Equity Investments. (21 minutes)**

Martin Gomez holds 100 shares of each of the stocks in a price-weighted index and reinvests cash dividends in additional shares. Assuming there are no stock splits, stock dividends, or changes in the makeup of the index, how will Gomez's portfolio return compare with the price return of the index if the low-priced index stocks outperform the high-priced index stocks?

- A) Gomez's portfolio return will be higher.**
- B) The price return of the index will be higher.**
- C) Gomez's portfolio return will be equal to the price return of the index.**

#### Explanation

The price return on the index does not include cash dividends. Since the reinvested dividends will add to the number of shares of those stocks that pay dividends, Gomez's portfolio return (total return) will be higher than the price return on the index. The relative performance of high-priced and low-priced stocks does not affect this result.

#### **For Further Reference:**

(Study Session 12, Module 37.1, LOS 37.b)

CFA® Program Curriculum, Volume 4, page 281

---

### Question #74 of 120

Question ID: 1207455

Evelyn Stram, CFA, places a good-till-cancelled limit buy order at 86 for a stock. Stram's order specifies:

- A) clearing and validity instructions.**
- B) validity and execution instructions.**
- C) execution and clearing instructions.**

#### Explanation

Good-till-cancelled is a validity instruction, which indicates when an order may be filled. Execution instructions include limit orders and market orders, as well as instructions regarding trade size and visibility. A clearing instruction indicates how to arrange final settlement of the trade.

#### **For Further Reference:**

(Study Session 12, Module 36.3, LOS 36.g)

CFA® Program Curriculum, Volume 4, page 246

---

### Question #75 of 120

Question ID: 1207462

The type of equity depository receipt that gives its owners the right to vote and receive dividends from a company's shares is *best* described as:

- A) a global depository receipt.
- B) a sponsored depository receipt.**
- C) a fully-owned depository receipt.

Explanation

The owner of a sponsored DR share has the same voting rights and receives the same dividends as the owner of a common share of the firm. With an unsponsored DR, the depository bank retains the voting rights. A global depository receipt may be sponsored or unsponsored.

**For Further Reference:**

(Study Session 13, Module 39.2, LOS 39.d)

CFA® Program Curriculum, Volume 4, page 365

---

**Question #76 of 120**

Question ID: 1207464

Ron Egan, CFA, classifies firms in the transportation industry in peer groups that include airlines and bus operators. Egan learns that one of the airlines, Acme, derives half its revenue from its Acme Bus Lines subsidiary. Egan adds Acme to his peer group for bus operators while continuing to include Acme in his peer group for airlines. Is Egan's treatment of Acme appropriate?

- A) Yes.**
- B) No, because each company should be included in only one peer group.
- C) No, because the bus operations are not the company's principal business activity.

Explanation

Peer groups should include comparable companies with similar business activities. An analyst can appropriately include a company in multiple peer groups if the company's business activities are comparable to firms in more than one peer group.

**For Further Reference:**

(Study Session 13, Module 40.1, LOS 40.d)

CFA® Program Curriculum, Volume 4, page 398

---

**Question #77 of 120**

Question ID: 1207466

An analyst gathered the following data about a company:

- A historical earnings retention rate of 60% that is projected to continue into the future.
- A sustainable return on equity of 10%.
- A beta of 1.0.
- The nominal risk-free rate is 5%.
- The expected market return is 10%.

If next year's EPS is \$2 per share, what value should be estimated for this stock?

A) \$20.00.

B) \$30.50.

C) \$35.45.

Explanation

dividend payout = 1 – earnings retention rate = 1 – 0.6 = 0.4

$$R_S = R_f + \beta (R_{mkt} - R_f) = 0.05 + 1.0 (0.10 - 0.05) = 0.10$$

$g = (\text{retention rate})(\text{ROE}) = (0.6)(0.10) = 0.06$

$$P/E = \frac{\text{dividend payout}}{k-g} = \frac{0.4}{0.10-0.06} = 10$$

price = (E)(P/E) = (2)(10) = \$20

Alternatively, dividend will be 40% of \$2 earnings or \$0.80, and  $\frac{0.80}{0.10-0.06} = \$20$ .

**For Further Reference:**

(Study Session 13, Module 41.2, LOS 41.g)

CFA® Program Curriculum, Volume 4, page 459

---

**Question #78 of 120**

Question ID: 1207458

The type of equity index *most likely* to require rebalancing is:

A) a price-weighted index.

B) an equal-weighted index.

C) a market-capitalization index.

Explanation

The individual security weights in an equal-weighted index must be rebalanced periodically to restore equal weights as security prices change over time. The weights in a price-weighted index are determined by securities prices. Market capitalization-weighted indexes are self-adjusting to a large extent.

**For Further Reference:**

(Study Session 12, Modules 37.1, 37.2, LOS 37.c, 37.f, 37.k)

CFA® Program Curriculum, Volume 4, page 284

CFA® Program Curriculum, Volume 4, page 293

CFA® Program Curriculum, Volume 4, page 305

---

**Question #79 of 120**

Question ID: 1207459

A high yield bond fund states that through active management, the fund's return has outperformed an index of Treasury securities by 4% on average over the past five years. As a performance benchmark for this fund, the index chosen is:

- A) appropriate.
- B) inappropriate, because the index return does not reflect active management.
- C) inappropriate, because the index does not reflect the actual bonds in which the fund invests.**

Explanation

Security market indexes may be used as benchmarks for the performance of active managers, but the index chosen should represent the universe of securities from which the manager is choosing. Here, an index of high yield bonds would be a more appropriate benchmark.

**For Further Reference:**

(Study Session 12, Module 37.2, LOS 37.f)

CFA® Program Curriculum, Volume 4, page 293

---

**Question #80 of 120**

Question ID: 1207463

The change in the intrinsic value of a firm's common stock resulting from an increase in ROE *most likely*:

- A) increases the stock's intrinsic value.
- B) decreases the stock's intrinsic value.
- C) depends on the reason for the increase in ROE.**

Explanation

While an increase in a firm's ROE due to a sharp increase in earnings will, if unexpected, lead to an increase in the intrinsic value of its shares, an increase in a firm's ROE due to the repurchase of stock with debt will not necessarily increase the intrinsic value of the firm's shares, as any increase in ROE may be offset by an increase in the risk inherent in the firm's shares.

**For Further Reference:**

(Study Session 13, Module 39.2, LOS 39.h)

CFA® Program Curriculum, Volume 4, page 373

---

**Question #81 of 120**

Question ID: 1207461

Which of the following statements about short selling is *least accurate*?

- A) A short seller is required to set up a margin account.
- B) A short sale involves securities the investor does not own.
- C) A short seller loses if the price of the stock sold short decreases.**

Explanation

The short seller loses if the stock price increases. The other choices are accurate statements.

**For Further Reference:**

(Study Session 12, Module 38.1, LOS 38.e)

**Question #82 of 120**

Question ID: 1207460

Compared to an index of 100 U.S. exchange-traded stocks, an index of 100 U.S. government and corporate bonds will *most likely*:

- A) reflect equally timely price data.
- B) be more difficult to build and maintain.**
- C) have less turnover among the securities in the index.

Explanation

Bond indexes are more difficult to build and maintain than stock indexes for several reasons. Bonds in an index have to be replaced as they mature, so turnover is likely to be greater in a bond index than in a stock index. Many bonds lack the continuous trade data that exists for exchange-traded equities.

**For Further Reference:**

(Study Session 12, Module 37.2, LOS 37.i)

CFA® Program Curriculum, Volume 4, page 300

**Question #83 of 120**

Question ID: 1207467

Beth Knight, CFA, and David Royal, CFA, are independently analyzing the value of Bishop, Inc., stock. Bishop paid a dividend of \$1 last year. Knight expects the dividend to grow by 10% in each of the next three years, after which it will grow at a constant rate of 4% per year. Royal also expects a temporary growth rate of 10% followed by a constant growth rate of 4%, but he expects the supernormal growth to last for only two years. Knight estimates that the required return on Bishop stock is 9%, but Royal believes the required return is 10%. Royal's valuation of Bishop stock is approximately:

- A) equal to Knight's valuation.
- B) \$5 less than Knight's valuation.**
- C) \$5 greater than Knight's valuation.

Explanation

You can select the correct answer without calculating the share values. Royal is using a shorter period of supernormal growth and a higher required rate of return on the stock. Both of these factors will contribute to a lower value using the multistage DDM.

$$\text{Knight: } \frac{\$1(1.10)}{1.09} + \frac{\$1(1.10)^2}{1.09^2} + \frac{\$1(1.10)^3 / (0.09 - 0.04)}{1.09^2} = \$24.43$$

$$\text{Royal: } \frac{\$1(1.10)}{1.10} + \frac{\$1(1.10)^2 / (0.10 - 0.04)}{1.10} = \$19.33$$

Royal's valuation is \$5.10 less than Knight's valuation.

**For Further Reference:**

(Study Session 13, Module 41.2, LOS 41.g)

CFA® Program Curriculum, Volume 4, page 459

## Question #84 of 120

Question ID: 1207465

An industry in the growth phase of the industry life cycle is *most likely* to experience:

- A) increasing prices.
- B) increasing profitability.**
- C) intense competition among competitors.

### Explanation

An industry in the growth stage is usually characterized by increasing profitability, decreasing prices, and a low degree of competition among competitors.

### **For Further Reference:**

(Study Session 13, Module 40.2, LOS 40.h)

CFA® Program Curriculum, Volume 4, page 414

---

## Question #85 of 120

Question ID: 1207468

Acquire Corp. has a business model based on making accretive acquisitions each year. The company has historically been successful in implementing its strategy. Earnings per share have grown each of the last five years at a 15% compounded rate. Acquire's two primary business segments are engineering construction and mining. During the past year, Acquire purchased a services company with large net operating losses. The purchase price was one-half the company's current market value. The *most appropriate* technique to value Acquire is based on its:

- A) price-to-book value ratio.
- B) forward price-to-earnings ratio.**
- C) trailing price-to-sales ratio.

### Explanation

The new acquisition has likely changed the nature of Acquire's business so that historical information is not as relevant to the investment decision-making process. The most appropriate earnings multiple for analyzing Acquire is the forward price-to-earnings ratio.

### **For Further Reference:**

(Study Session 13, Module 41.3, LOS 41.i)

CFA® Program Curriculum, Volume 4, page 468

---

## Question #86 of 120

Question ID: 1207456

Liquidity is generally supplied by dealers in:

- A) a brokered market.
- B) an order-driven market.
- C) a quote-driven market.**

### Explanation

Liquidity is generally supplied by dealers in a quote-driven market, by other traders in an order-driven market, and by brokers (who arrange the trades) and traders in a brokered market.

**For Further Reference:**

(Study Session 12, Module 36.3, LOS 36.j)

CFA® Program Curriculum, Volume 4, page 256

---

**Question #87 of 120**

Question ID: 1212013

**Questions 87 through 99 relate to Fixed Income. (19.5 minutes)**

Consider the following Treasury spot rates expressed as bond equivalent yields:

| Maturity  | Spot Rate |
|-----------|-----------|
| 6 months  | 3.0%      |
| 1 year    | 3.5%      |
| 1.5 years | 4.0%      |
| 2 years   | 4.5%      |

If a Treasury note with two years remaining to maturity has a 5% semiannual coupon and is priced at \$1,008, the note is:

- A) overpriced.
- B) underpriced.**
- C) correctly priced.

Explanation

The market value of the Treasury note is the present value of the remaining coupons plus the present value of the principal, discounted at the semiannual rates available from dividing each annual spot rate in the table by two.

$$\begin{aligned}\text{value of T-note} &= \frac{\$25}{(1.015)} + \frac{\$25}{(1.0175)^2} + \frac{\$25}{(1.02)^3} + \frac{\$1,025}{(1.0225)^4} \\ &= \$1,010.05\end{aligned}$$

At \$1,008, the T-note is priced below the present value of its cash flows (\$1,010) and is therefore underpriced.

**For Further Reference:**

(Study Session 14, Module 44.2, LOS 44.c)

CFA® Program Curriculum: Volume 5, page 117

---

**Question #88 of 120**

Question ID: 1212020

Which of the following is an advantage of a callable bond (compared to an identical option-free bond) to an investor?

- A) Less reinvestment risk.
- B) Higher yield.**
- C) More convexity.

### Explanation

An issuer of a callable bond must compensate the bondholder when the issue is sold by offering a higher coupon rate or accepting a lower price than if the call feature was not included. Convexity will typically be much less than for an option-free bond, and reinvestment risk is greater for callable bonds.

### **For Further Reference:**

(Study Session 14, Module 42.2, LOS 42.f, Study Session 15, Module 46.3, LOS 46.h)

CFA® Program Curriculum: Volume 5, page 37

CFA® Program Curriculum: Volume 5, page 269

---

## **Question #89 of 120**

Question ID: 1212018

The bonds of Grinder Corp. trade at a G-spread of 150 basis points above comparable maturity U.S. Treasury securities. The option adjusted spread (OAS) on the Grinder bonds is 75 basis points. Using this information, and assuming that the Treasury yield curve is flat:

- A) the option cost is 75 basis points.**
- B) the zero-volatility spread is 75 basis points.**
- C) the zero-volatility spread is 225 basis points.**

### Explanation

The option cost is the difference between the zero volatility spread and the OAS, or  $150 - 75 = 75$  bp. With a flat yield curve, the G-spread and zero volatility spread will be the same.

### **For Further Reference:**

(Study Session 14, Module 44.5, LOS 44.k)

CFA® Program Curriculum: Volume 5, page 147

---

## **Question #90 of 120**

Question ID: 1212022

The yield spreads between corporate bonds and government bonds are *most likely* to decrease if:

- A) liquidity decreases in the market for the corporate bonds.**
- B) a credit rating downgrade on the corporate bonds becomes more likely.**
- C) investors increase their estimates of the recovery rate on the corporate bonds.**

### Explanation

Yield spreads reflect the credit quality of bond issuers and the liquidity of the market for their bonds. Narrowing (decreasing) yield spreads reflect improving credit quality or more liquidity. Widening (increasing) yield spreads reflect deteriorating credit quality or less liquidity. Increased estimates of the recovery rate in the event of default represent an improvement in investors' assessment of the issuer's credit quality and are likely to narrow yield spreads on the issuer's bonds.

### **For Further Reference:**

(Study Session 15, Module 47.1, LOS 47.a)

### Question #91 of 120

Question ID: 1212014

The full price of a bond:

- A) includes accrued interest.**
- B) includes commissions and taxes.
- C) is also known as the “clean” price.

#### Explanation

The full price is clean price plus accrued interest.

#### **For Further Reference:**

(Study Session 14, Module 44.2, LOS 44.d)

CFA® Program Curriculum: Volume 5, page 119

---

### Question #92 of 120

Question ID: 1212011

Which of the following is a disadvantage to bondholders if a bond has a sinking fund provision?

- A) Lower credit quality.**
- B) Unfavorable tax status.
- C) Greater reinvestment risk.**

#### Explanation

Reinvestment risk is a disadvantage of a sinking fund provision. Some bondholders will be repaid the bond principal earlier than the maturity date. If interest rates have declined since they bought the bonds, these bondholders will only be able to reinvest the returned principal at a lower rate of return. A sinking fund provision increases the credit quality of an issue and typically does not affect a bond's taxable status.

#### **For Further Reference:**

(Study Session 14, Module 42.2, LOS 42.e)

CFA® Program Curriculum: Volume 5, page 26

---

### Question #93 of 120

Question ID: 1212023

A debt covenant designates one of a holding company's subsidiaries as restricted. Which of the following credit-related considerations does this covenant address?

- A) Credit migration risk.**
- B) Structural subordination.**
- C) Payments to equity holders.

### Explanation

Restricted subsidiaries are those whose cash flows and assets are designated to service the debt of their holding company. Classifying a subsidiary as restricted alleviates structural subordination by making holding company debt rank pari passu with the subsidiary's debt.

### **For Further Reference:**

(Study Session 15, Module 47.2, LOS 47.j)

CFA® Program Curriculum: Volume 5, page 347

---

## **Question #94 of 120**

Question ID: 1212016

Consider a collateralized mortgage obligation (CMO) structure with one planned amortization class (PAC) class and one support tranche outstanding. If the prepayment speed is higher than the upper collar on the PAC:

- A) the life of the PAC tranche will increase.
- B) the PAC tranche has no risk of prepayments.
- C) the life of the support tranche will decrease.

### Explanation

If the prepayment speed is higher than the PAC collar, the support tranche receives more prepayments. The life of the support tranche will shorten. The PAC tranche *could* receive higher prepayments if the support tranche principal is fully repaid (i.e., a broken PAC). In this case, the support tranche is still outstanding, which means that hasn't happened yet.

### **For Further Reference:**

(Study Session 14, Modules 45.2, 45.3, LOS 45.e, 45.f)

CFA® Program Curriculum: Volume 5, page 194

CFA® Program Curriculum: Volume 5, page 194

---

## **Question #95 of 120**

Question ID: 1212021

For a bond currently priced at \$1,018 with an effective duration of 7.48, if the market yield moved down 75 basis points, the new price would be approximately:

- A) \$961.
- B) \$1,075.
- C) \$1,094.

### Explanation

$$\% \Delta \text{price} \approx -7.48(-0.0075) = 0.0561$$

$$\$1,018(1 + 0.0561) = \$1,075.11$$

### **For Further Reference:**

(Study Session 15, Module 46.3, LOS 46.i)

CFA® Program Curriculum: Volume 5, page 270

---

**Question #96 of 120**

Question ID: 1212015

A public offering of bonds issued over a period of time is *most accurately* described as:

- A) a serial structure.
- B) a shelf registration.**
- C) a waterfall structure.

Explanation

In a shelf registration, an entire issue is registered with securities regulators but the bonds are sold to the public over a period of time as the issuer needs to raise funds. In a serial bond issue, bonds with multiple maturity dates are issued at the same time. A waterfall structure is issued in tranches with differing priority of claims.

**For Further Reference:**

(Study Session 14, Modules 43.1, 43.2, LOS 43.c, 43.g)

CFA® Program Curriculum: Volume 5, page 66

CFA® Program Curriculum: Volume 5, page 79

---

**Question #97 of 120**

Question ID: 1212017

The minimum data required to calculate the implied forward rate for three years beginning three years from now is:

- A) the 3-year and 6-year spot rates.**
- B) the 4-year, 5-year, and 6-year spot rates.
- C) spot rates at 1-year intervals for the 6-year period.

Explanation

If we want the 3-year forward rate in three years, the appropriate formula is:

$${}_{3y}3y = \left[ \frac{(1+S_6)^6}{(1+S_3)^3} \right]^{1/3} - 1$$

$S_6$  = 6-year spot rate and  $S_3$  = 3-year spot rate.

**For Further Reference:**

(Study Session 14, Module 44.4, LOS 44.j)

CFA® Program Curriculum: Volume 5, page 143

---

**Question #98 of 120**

Question ID: 1212019

A 3-year, 6% coupon, semiannual-pay note has a yield to maturity of 5.5%. If an investor holds this note to maturity and earns a 4.5% return on reinvested coupon income, his realized yield on the note is *closest* to:

- A) 5.46%.**

- B)** 5.57%.
- C)** 5.68%.

Explanation

This question does not require calculations. Because the return on reinvested coupon interest is less than the note's yield to maturity, the investor's realized yield on the note must be less than the YTM. Only Choice A can be correct.

**For Further Reference:**

(Study Session 15, Module 46.1, LOS 46.a)

CFA® Program Curriculum: Volume 5, page 241

---

**Question #99 of 120**

Question ID: 1212012

An annual-pay, 4% coupon, 10-year bond has a yield to maturity of 5.2%. If the price of this bond is unchanged two years later, its yield to maturity at that time is:

- A)** 5.2%.
- B)** less than 5.2%.
- C)** greater than 5.2%.

Explanation

This bond is priced at a discount to par value because its 4% coupon is less than its 5.2% yield to maturity. As the bond gets closer to maturity, the discount will amortize toward par value, which means its price will increase if its yield remains unchanged. For its price to remain unchanged, its yield would have to increase.

Price with 10 years to maturity:

$$N = 10; I/Y = 5.2; PMT = 40; FV = 1,000; CPT PV = -908.23$$

Yield with 8 years to maturity:

$$N = 8; PMT = 40; FV = 1,000; PV = -908.23; CPT I/Y = 5.446\%$$

**For Further Reference:**

(Study Session 14, Module 44.3, LOS 44.b)

CFA® Program Curriculum: Volume 5, page 113

---

**Question #100 of 120**

Question ID: 1212024

**Questions 100 through 106 relate to Derivatives. (10.5 minutes)**

Ed Verdi has a long position in a European put option on a stock. At expiration, the stock price is greater than the exercise price. The value of the put option to Verdi on its expiration date is:

- A)** zero.
- B)** positive.
- C)** negative.

### Explanation

At expiration, the value to the holder (long position) of a put option on a stock is the greater of zero or the exercise price minus the stock price. If the stock price is greater than the exercise price, the value of a put option to the holder is zero and the holder will allow the option to expire unexercised.

### **For Further Reference:**

(Study Session 16, Module 48.2, LOS 48.d)

---

## **Question #101 of 120**

Question ID: 1207485

A company is *most likely* to enter a plain vanilla interest rate swap as the fixed-rate payer if its balance sheet has predominantly:

- A) floating-rate assets, and management expects interest rates to increase.
- B) fixed-rate liabilities, and management expects interest rates to decrease.
- C) floating-rate liabilities, and management expects interest rates to increase.**

### Explanation

If interest rates are expected to increase, a company with predominantly floating-rate liabilities could benefit from entering an interest rate swap on the pay-fixed, receive-floating side. An increase in interest rates will increase the floating-rate payments received on the swap, offsetting the increase in the required payments on the floating-rate liability.

### **For Further Reference:**

(Study Session 16, Modules 48.2, 49.2, LOS 48.c, 49.i)

CFA® Program Curriculum, Volume 5, page 394

---

## **Question #102 of 120**

Question ID: 1207484

Which of the following is *most likely* to be an impediment to arbitrage?

- A) The law of one price.
- B) Short-sale restrictions.**
- C) Investors' risk aversion.

### Explanation

Restrictions on short sales may be an impediment to arbitrage when a security is overpriced relative to another security or derivative that has identical cash flows. Because arbitrage transactions are theoretically riskless, risk aversion should not be viewed as an impediment. The law of one price is the theory arbitrage trading will force two assets with the same cash flows toward the same price.

### **For Further Reference:**

(Study Session 16, Modules 48.2, 49.1, LOS 48.e, 49.a)

CFA® Program Curriculum, Volume 5, page 430

CFA® Program Curriculum, Volume 5, page 456

---

## Question #103 of 120

Question ID: 1207487

The put-call-forward parity relationship is similar to the standard put-call parity relationship with a forward price substituted for:

- A) the risk-free bond.
- B) the underlying asset.**
- C) either the call or put option.

### Explanation

The put-call-forward parity relationship is the same as the standard put-call parity relationship, with the present value of the forward price substituted for the underlying asset.

### **For Further Reference:**

(Study Session 16, Module 49.3, LOS 49.I, 49.m)

CFA® Program Curriculum, Volume 5, page 491

CFA® Program Curriculum, Volume 5, page 494

---

## Question #104 of 120

Question ID: 1207482

Over-the-counter derivatives are:

- A) standardized and backed by a clearinghouse.
- B) not standardized but are backed by a clearinghouse.
- C) neither standardized nor backed by a clearinghouse.**

### Explanation

Over-the-counter derivatives are customized private contracts between counterparties.

### **For Further Reference:**

(Study Session 16, Module 48.1, LOS 48.a)

CFA® Program Curriculum, Volume 5, page 386

---

## Question #105 of 120

Question ID: 1212025

The time value of an option is *most accurately* described as:

- A) increasing as the option approaches its expiration date.
- B) equal to the entire premium for an out-of-the-money option.**
- C) the amount by which the intrinsic value exceeds the option premium.

### Explanation

The price (or premium) of an option is its intrinsic value plus its time value. An out-of-the-money option has an intrinsic value of zero, so its entire premium consists of time value. Time value is zero at an option's expiration date. Time value is the amount by which an option's premium exceeds its intrinsic value.

**For Further Reference:**

(Study Session 16, Module 49.3, LOS 49.j, 49.k)

CFA® Program Curriculum, Volume 5, page 481

CFA® Program Curriculum, Volume 5, page 482

---

**Question #106 of 120**

Question ID: 1207486

A 60-day forward rate agreement (FRA) on 60-day U.S. dollar LIBOR has a fixed rate of 6%. If, at the initiation of the contract, 60-day LIBOR is 5%, the payment in 60 days:

- A) is unknown.**
- B) will be received by the long position.**
- C) will be received by the short position.**

Explanation

The payment on an FRA is unknown until the settlement date of the forward, when the future 60-day rate is known.

**For Further Reference:**

(Study Session 16, Module 49.2, LOS 49.f)

CFA® Program Curriculum, Volume 5, page 473

---

**Question #107 of 120**

Question ID: 1212027

**Questions 107 through 113 relate to Alternative Investments. (10.5 minutes)**

Over time, compared to traditional stock and bond investments, the commodities asset class has exhibited:

- A) lower returns and lower price volatility.**
- B) lower returns and higher price volatility.**
- C) higher returns and lower price volatility.**

Explanation

Returns on commodities over time have been lower than returns on global stocks or bonds, and price volatility has been higher.

**For Further Reference:**

(Study Session 17, Module 50.2, LOS 50.b)

---

**Question #108 of 120**

Question ID: 1212028

A hedge fund uses derivative positions to take a long position in the Japanese yen and a short position in the euro. The classification of this hedge fund is *most likely*:

- A) an event-driven fund.**

- B) a macro strategy fund.**
- C) a quantitative directional fund.**

Explanation

Macro strategy funds invest based on expected shifts of global economies, primarily in currency and interest rate derivatives. Quantitative directional funds take long and short positions in equities based on technical analysis. Event-driven funds seek to profit from investment strategies based on specific corporate events and one-time transactions.

**For Further Reference:**

(Study Session 17, Module 50.2, LOS 50.b)

---

**Question #109 of 120**

Question ID: 1212029

Compared to its net asset value (NAV) calculated in accordance with accounting standards, a hedge fund's trading NAV:

- A) will be lower because of adjustments for illiquid positions.**
- B) is likely to be higher because of upward bias in model-based security values.**
- C) may be higher or lower, reflecting gains or losses on securities designated for short-term trading.**

Explanation

If a fund calculates a trading NAV, it will adjust market prices downward for securities in which it holds positions that are large relative to trading volume or total value outstanding and thus are less liquid.

**For Further Reference:**

(Study Session 17, Module 50.2, LOS 50.e)

CFA® Program Curriculum, Volume 6, page 10

---

**Question #110 of 120**

Question ID: 1212026

The mezzanine financing portion of a leveraged buyout (LBO) is *most likely* to:

- A) represent committed capital.**
- B) be convertible to equity or include warrants.**
- C) have seniority over other bonds issued to finance the LBO.**

Explanation

In the context of an LBO, mezzanine financing refers to debt that carries warrants or equity conversion features. This debt is typically subordinated to other bonds that are issued to finance the LBO. Committed capital is the investment of limited partners in a private equity fund and does not include debt that the fund issues to finance a particular LBO.

**For Further Reference:**

(Study Session 17, Module 50.2, LOS 50.b)

---

## Question #111 of 120

Question ID: 1207490

A characteristic of alternative investments that distinguishes them from traditional investments is that:

- A) redemptions are more restricted.**
- B) they are subject to stricter regulations.
- C) managers hold more diversified portfolios.

### Explanation

Greater restrictions on investor redemptions are a characteristic of alternative investments. Alternative investments are less regulated than traditional investments and managers tend to hold more concentrated portfolios.

### **For Further Reference:**

(Study Session 17, Module 50.1, LOS 50.a)

CFA® Program Curriculum, Volume 6, page 6

---

## Question #112 of 120

Question ID: 1212030

Public-private partnerships are *most likely* to be a vehicle for investing in:

- A) tangible collectibles.
- B) distressed securities.
- C) greenfield infrastructure.**

### Explanation

Public-private partnerships are a vehicle for investing in infrastructure.

### **For Further Reference:**

(Study Session 17, Module 50.1, LOS 50.b)

---

## Question #113 of 120

Question ID: 1207495

When conducting due diligence on a vehicle for alternative investments, which of the following observations would *most likely* be positive for a potential investor?

- A) Auditing and reporting to investors are performed by reliable third parties.**
- B) The general partner is prohibited from commingling his own funds with the limited partners' funds.
- C) A well-known and highly regarded investor holds shares that represent 20% of the vehicle's assets under management.

### Explanation

Use of quality third-party service providers by an investment vehicle is a positive sign for potential investors. Manager and investor interests are more likely to be aligned if the general partner invests alongside the limited partners. If a single investor represents a large

portion of assets under management, and that investor chooses to withdraw, the manager may be forced to liquidate assets at unfavorable prices.

**For Further Reference:**

(Study Session 17, Module 50.2, LOS 50.f)

CFA® Program Curriculum, Volume 6, page 57

---

**Question #114 of 120**

Question ID: 1207497

**Questions 114 through 120 relate to Portfolio Management. (10.5 minutes)**

Which of the following types of institutions is *most likely* to have the lowest risk tolerance?

- A) Commercial bank.**
- B) College endowment.**
- C) Mutual fund company.**

Explanation

Banks typically need to maintain excess reserves in order to meet regulatory requirements. As a result, a bank must invest in assets that are more conservative than those invested by other types of financial institutions. An endowment will usually have significant long-term spending requirements in addition to its current expenses. Thus, a college endowment should accept higher risk in order to attain the returns indicated by its mandate. A mutual fund company may invest in many types of securities depending on the type of funds being managed. Investments may range from conservative money market funds to more aggressive derivatives. Therefore, a mutual fund company's level of risk tolerance may be greater than or less than those of a bank or endowment.

**For Further Reference:**

(Study Session 18, Module 52.1, LOS 52.c)

CFA® Program Curriculum, Volume 6, page 133

---

**Question #115 of 120**

Question ID: 1207496

Under which type of pension plan are retirement benefit payments an obligation of the sponsoring firm?

- A) Defined benefit plan only.**
- B) Defined contribution plan only.**
- C) Both a defined benefit plan and a defined contribution plan.**

Explanation

Promised retirement payments from a defined benefit pension plan are an obligation of the sponsoring firm. A defined contribution plan does not promise a specific periodic payment after retirement and the firm's obligations are limited to its promised contributions to current employees' accounts.

**For Further Reference:**

(Study Session 18, Module 51.1, LOS 51.c)

CFA® Program Curriculum, Volume 6, page 93

---

**Question #116 of 120**

Question ID: 1207498

Which of the following possible portfolios is *least likely* to lie on the efficient frontier?

| Portfolio | Expected Return | Standard Deviation |
|-----------|-----------------|--------------------|
| X         | 9%              | 12%                |
| Y         | 11%             | 10%                |
| Z         | 13%             | 15%                |

- A) Portfolio X.
- B) Portfolio Y.
- C) Portfolio Z.

Explanation

Portfolio X has a lower expected return and a higher standard deviation than Portfolio Y. X must be inefficient.

**For Further Reference:**

(Study Session 18, Module 52.3, LOS 52.h)

CFA® Program Curriculum, Volume 6, page 163

---

**Question #117 of 120**

Question ID: 1207499

In extending the 3-factor model of Fama and French, the additional factor suggested by Carhart that is often used is:

- A) GDP growth.
- B) **price momentum.**
- C) market-to-book value.

Explanation

In addition to the three factors of the Fama and French model, market-to-book, firm size, and excess returns on the market, Carhart added a momentum factor based on prior relative price performance.

**For Further Reference:**

(Study Session 18, Module 53.1, LOS 53.d)

CFA® Program Curriculum, Volume 6, page 206

---

**Question #118 of 120**

Question ID: 1212031

Marcia Kostner, CFA, is an advisor to individual investors. To determine each of her clients' risk tolerance objectively, Kostner uses a mathematical formula with inputs that include the client's age, family size, insurance coverage, liquidity, income, and net worth. What is the *most likely* shortcoming of Kostner's approach to assessing risk tolerance?

- A) Net worth is unrelated to an investor's risk tolerance.

- B) This approach does not consider the investor's attitude toward risk.**
- C) Treating clients differently based on their ages violates the Code and Standards.

Explanation

When determining an investor's risk tolerance, an advisor should analyze the investor's personal situation, but should also gauge the investor's attitude toward risk. Risk tolerance is affected by the investor's psychological profile (i.e., willingness to take risk) as well as by the investor's ability to take risk. Age is an important influence on risk tolerance; younger investors generally are more able to withstand short-term losses because they have a longer time horizon in which to recover. Investors with high net worth are also more able to withstand short-term losses than investors with lower net worth, and thus tend to be more tolerant of risk.

**For Further Reference:**

(Study Session 19, Module 54.1, LOS 54.c, 54.d)

CFA® Program Curriculum, Volume 6, page 250

CFA® Program Curriculum, Volume 6, page 252

**Question #119 of 120**

Question ID: 1207501

An analyst gathered the following data about three stocks:

| Stock | Beta | Estimated Return |
|-------|------|------------------|
| A     | 1.5  | 18.1%            |
| B     | 1.1  | 15.7%            |
| C     | 0.6  | 12.5%            |

If the risk-free rate is 8%, and the market risk premium is 7%, the analyst is *least likely* to recommend buying:

- A) Stock A.**
- B) Stock B.
- C) Stock C.

Explanation

Stock A:  $k_A = 8\% + 1.5(7\%) = 18.5\%$ . Because the estimated return of 18.1% is less than the required return of 18.5%, Stock A is *overvalued*.

Stock B:  $k_B = 8\% + 1.1(7\%) = 15.7\%$ . Because the estimated return of 15.7% equals the required return of 15.7%, Stock B is *properly valued*.

Stock C:  $k_C = 8\% + 0.6(7\%) = 12.2\%$ . Because the estimated return of 12.5% is greater than the required return of 12.2%, Stock C is *undervalued*.

**For Further Reference:**

(Study Session 18, Module 53.2, LOS 53.h)

CFA® Program Curriculum, Volume 6, page 217

**Question #120 of 120**

Question ID: 1207500

Which of the following equity securities is *most likely* to have a beta greater than one?

- A) Utility stock.
- B) Health care stock.
- C) Homebuilder stock.**

Explanation

Stocks of cyclical firms, such as homebuilders, tend to have high systematic risk (i.e., high beta). Stocks of noncyclical firms, such as utility or health care companies, tend to respond less to changes in systematic risk factors (i.e., they have low betas).

**For Further Reference:**

(Study Session 13, Module 40.1, LOS 40.c, 40.e, Study Session 18, Module 53.1, LOS 53.c, 53.e)

CFA® Program Curriculum, Volume 6, page 204

CFA® Program Curriculum, Volume 6, page 206

CFA® Program Curriculum, Volume 4, page 390